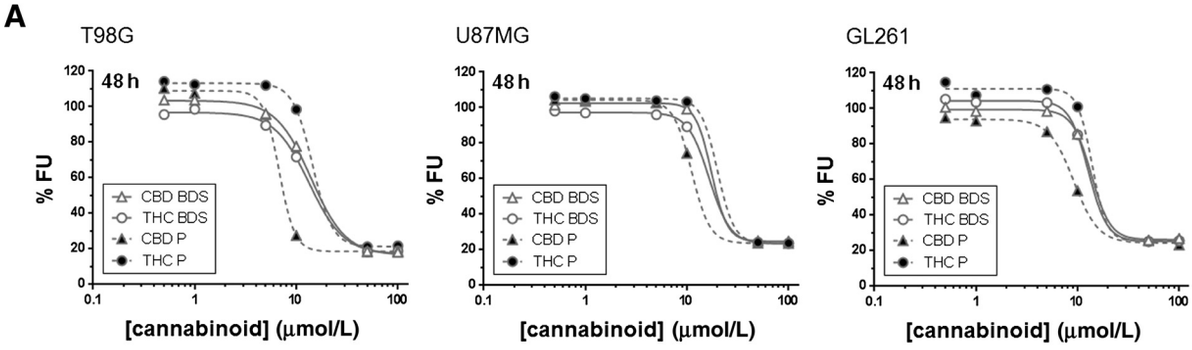

IC_{50} (μmol/L)	T98G			U87MG			GL261		
	24 h	48 h	72 h	24 h	48 h	72 h	24 h	48 h	72 h
CBD BDS	16 ± 2.3	16 ± 4.7	13 ± 2.6	14 ± 3.5	19 ± 4.0	11 ± 0.4	12 ± 1.7	12 ± 1.4	11 ± 0.4
THC BDS	12 ± 4.0	14 ± 2.8	10 ± 4.0	15 ± 3.0	17 ± 3.0	11 ± 0.7	13 ± 3.3	11 ± 0.8	10 ± 0.5
CBD P	10 ± 2.1	8.4 ± 2.9	5.2 ± 1.3	12 ± 3.4	14 ± 7.1	9.7 ± 1.3	11 ± 1.0	11 ± 4.2	10 ± 3.9
THC P	15 ± 2.1	16 ± 3.2	13 ± 2.7	16 ± 2.2	19 ± 3.9	14 ± 2.9	13 ± 2.9	14 ± 1.7	12 ± 1.5

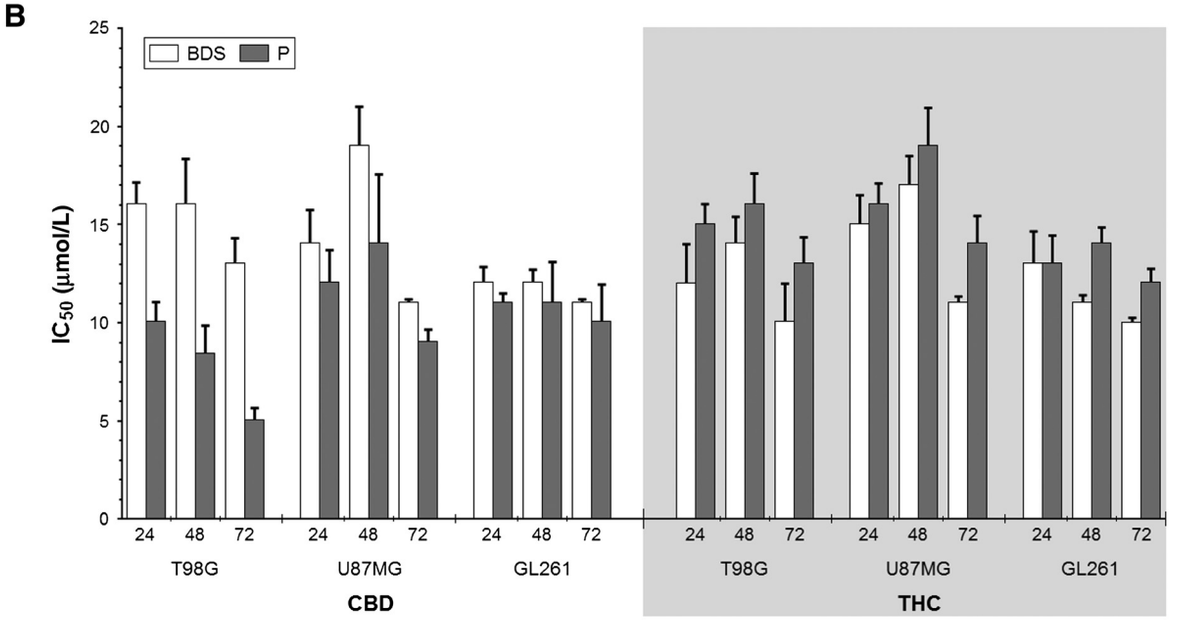

Figure 1. Effect of CBD and THC on cell proliferation and cell-cycle dynamics. T98G, U87MG, and GL261 were cultured with CBD or THC in both the P and BDS forms. Cell number was measured at 24, 48, and 72 hours, and the concentration required to reduce cell numbers by 50% (IC_{50}) were calculated using curve fitting software. The dose–response curves shown are from the 48-hour time point, where % FU represents the fraction of cells that were unaffected (A). Closer examination of the efficacy of the P and BDS forms of each cannabinoid revealed CBD was more active in the P form, whereas THC was more active in the BDS form (B). Each data point is the mean and SD of at least four separate experiments.

forms was tested and no significant differences were observed; however, combinations were typically more synergistic in U87MG cells than in T98G cells. In addition, combining CBD with THC did not alter the DNA profile of the cells (Fig. 2B).

CBD and THC alter intracellular signaling molecules

To investigate whether the effects that cannabinoids had on cell proliferation were associated with changes to intracellular signaling pathways, we analyzed lysates harvested from appropriately treated cells for the

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